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# From Legacy to Leading Edge

## Modernizing Data with Fabric

**Russ Seaman**
Solution Architect II | TD Synnex

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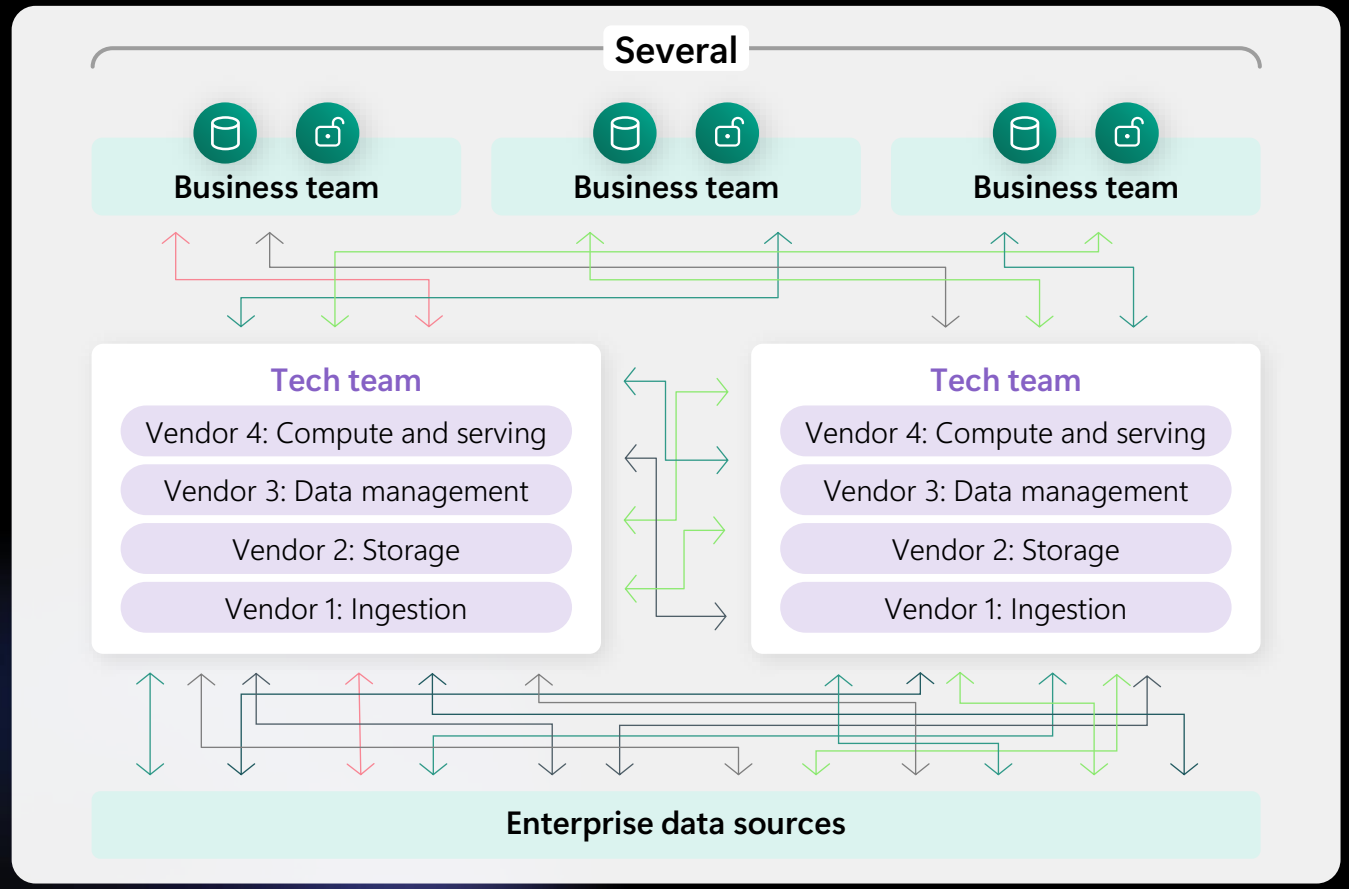
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# What is Fabric?



And more importantly, why would we want to use it?

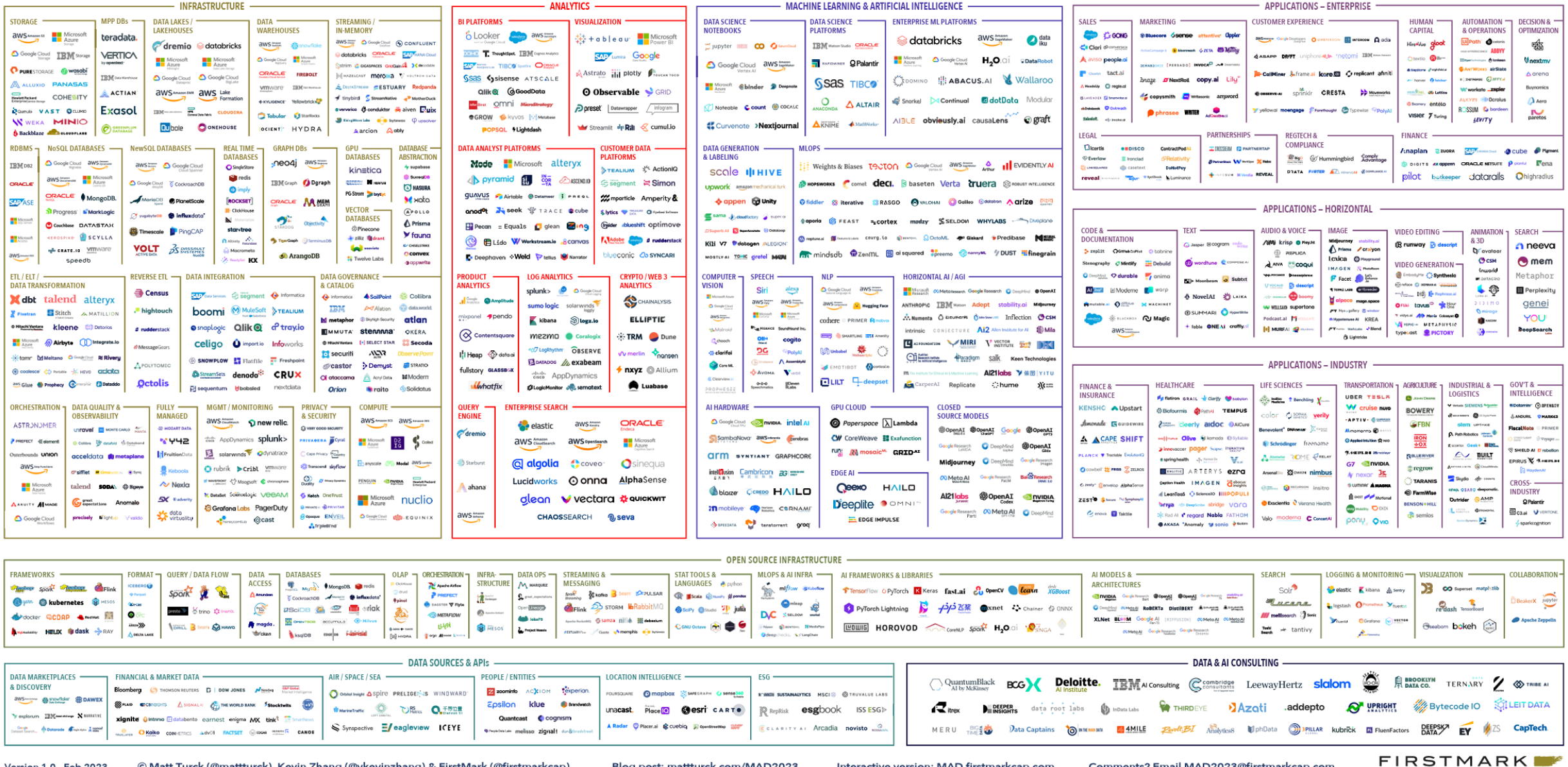
# The starting line




1. Data copies and infrastructure inefficiencies
2. Limited interoperability between vendor services
3. Data exposure risks

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Version 1.0 - Feb 2023 © Matt Turck (@mattturck), Kevin Zhang (@ykevinzhang) & FirstMark (@firstmarkcap) Blog post: mattturck.com/MAD2023 Interactive version: MAD.firstmarkcap.com Comments? Email MAD2023@firstmarkcap.com

FIRSTMARK EARLY STAGE VENTURE CAPITAL

| From | To |
| --- | --- |
| Multiple analytics services | Unified stack |
| Disconnected data sources | All the data in one place |
| Isolated application | Entire estate |
| Gen AI bolt on | Gen AI built in |

# Microsoft Fabric

- Data Factory
- Data Engineering
- Data Warehouse
- Data Science
- Real-Time Intelligence
- Power BI
- Partner & Industry workloads

Copilot in Fabric

OneLake

Microsoft Purview

## Unify your analytics on a complete, governed platform

Reduce the cost and effort of integration with a unified, secure, and governed platform



## Establish a trusted data foundation

Connect to any data from a single, multi-cloud data lake and use a single copy of data across engines

## Empower every business user

Empower everyone to uncover insights accessible data, easy-to-use tools, and visuals embedded apps they use everyday

## Fuel your AI innovation

Adopt a data platform infused with AI at every layer to help you get more done, faster

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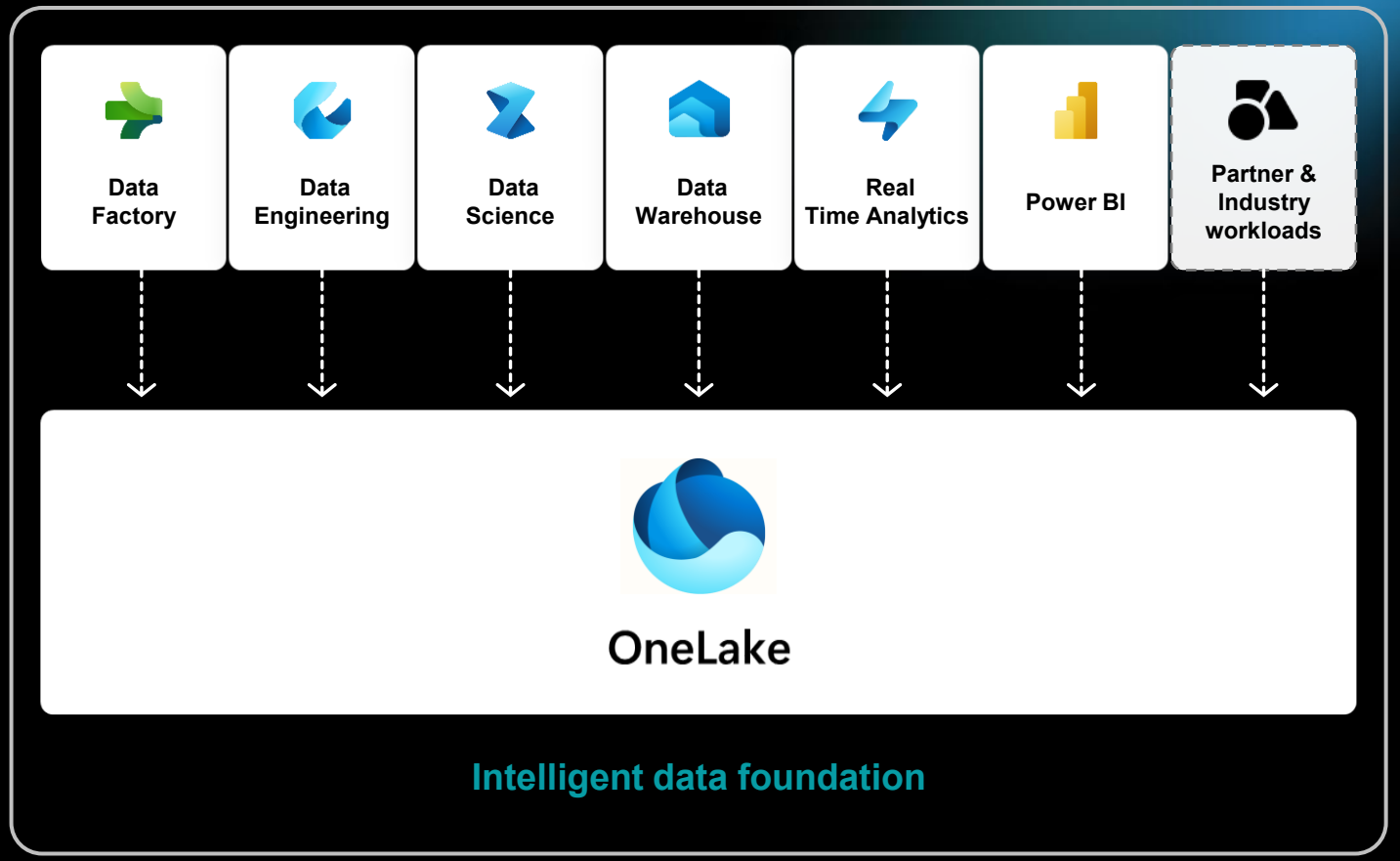
# OneLake to rule all your data.

It’s like “OneDrive”, just for your data.

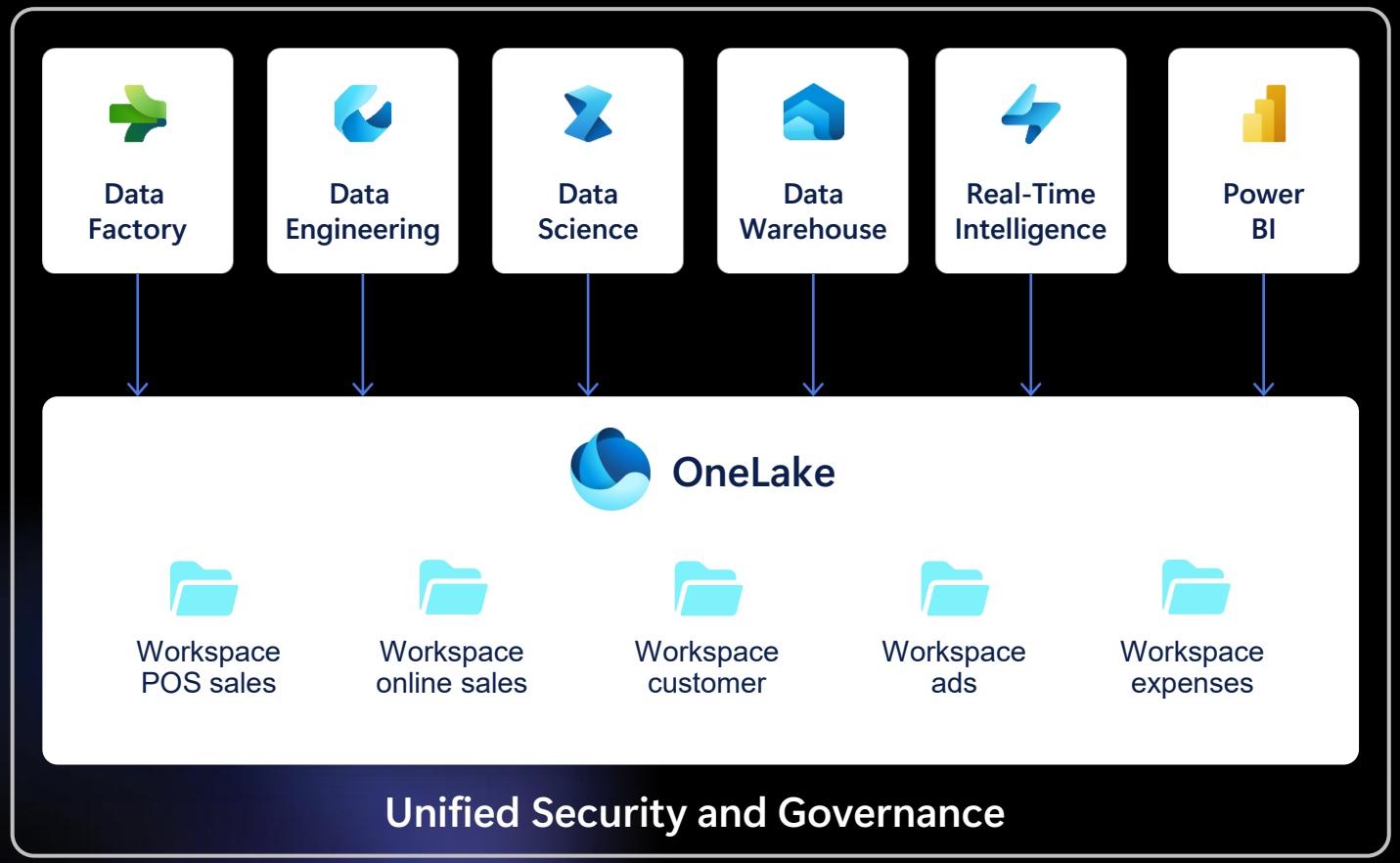


# OneLake for all your data

- A single SaaS lake for the whole organization
- Provisioned automatically with the tenant
- All workloads automatically store their data in the OneLake workspace folders
- All the data is organized in an intuitive hierarchical namespace
- The data in OneLake is automatically indexed for discovery, MIP labels, lineage, PII scans, sharing, governance and compliance



# A single unified SaaS data lake




Provisioned automatically with the tenant.

Any data in OneLake works with out-of-the-box governance such as data lineage, data protection, certification, catalog integration, etc. All data is ultimately under the control of a tenant admin.

OneLake enables distributed ownership. Different workspaces allow different parts of the organization to work independently while still contributing to the same data lake. Each workspace can have its own administrator, access control, region and capacity for billing.

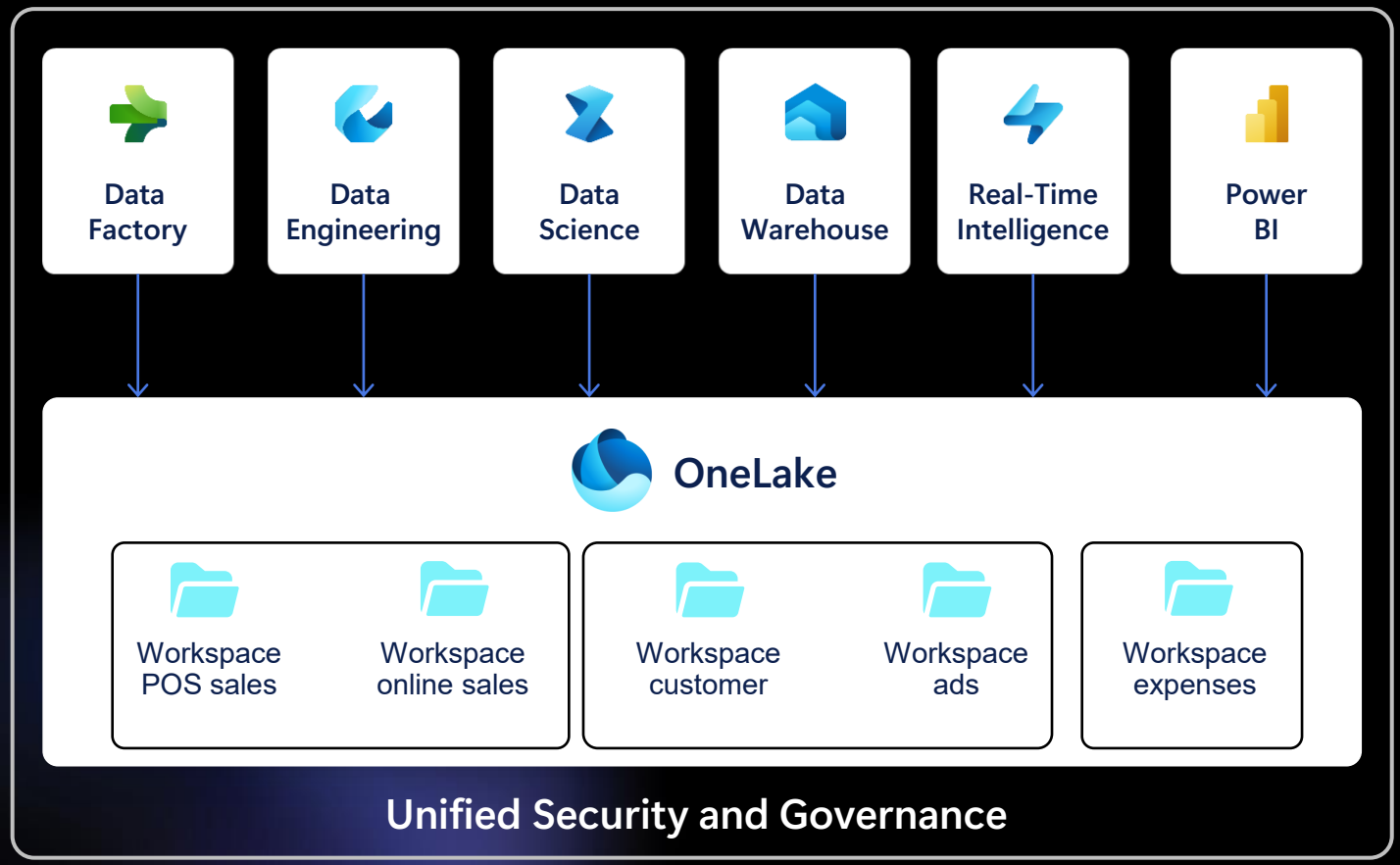
# OneLake for all domains

OneLake give a true data mesh as a service

Sales

Marketing

Finance




Introducing domains as an integral part of Fabric: A domain is a way to logically group together all the data in an organization relevant to an area or field, according to business needs.

Domains are defined with domain admins and contributors who can associate workspaces and group them together under a relevant domain.

Federated governance can be achieved by delegating settings to domain admins, thus allowing them to achieve more granular control over their business area.

Domains simplify discovery and consumption of data across the organization, thus allowing business optimized consumption.

Avoid data swamps by endorsing certain data as certified or promoted, thus encouraging reuse.

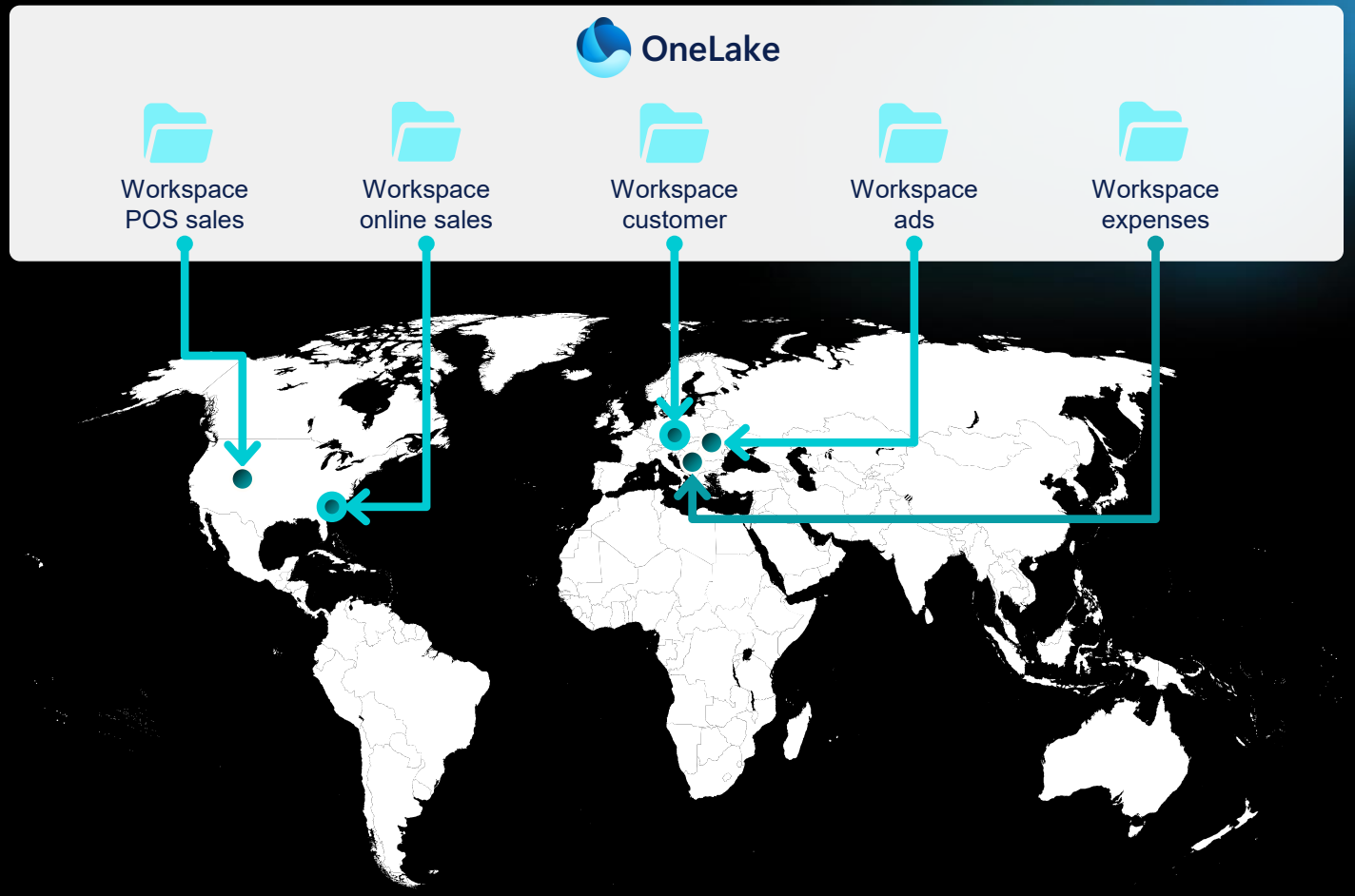
To achieve data residency requirements, workspaces can reside in different regions around the world while still being part of the same data lake.

Data can reside in different regions without the overhead of managing different storage resources and without creating data silos.

OneLake provisions storage resources for each workspace to meet demand for scale (capacity, throughput and IOPS).

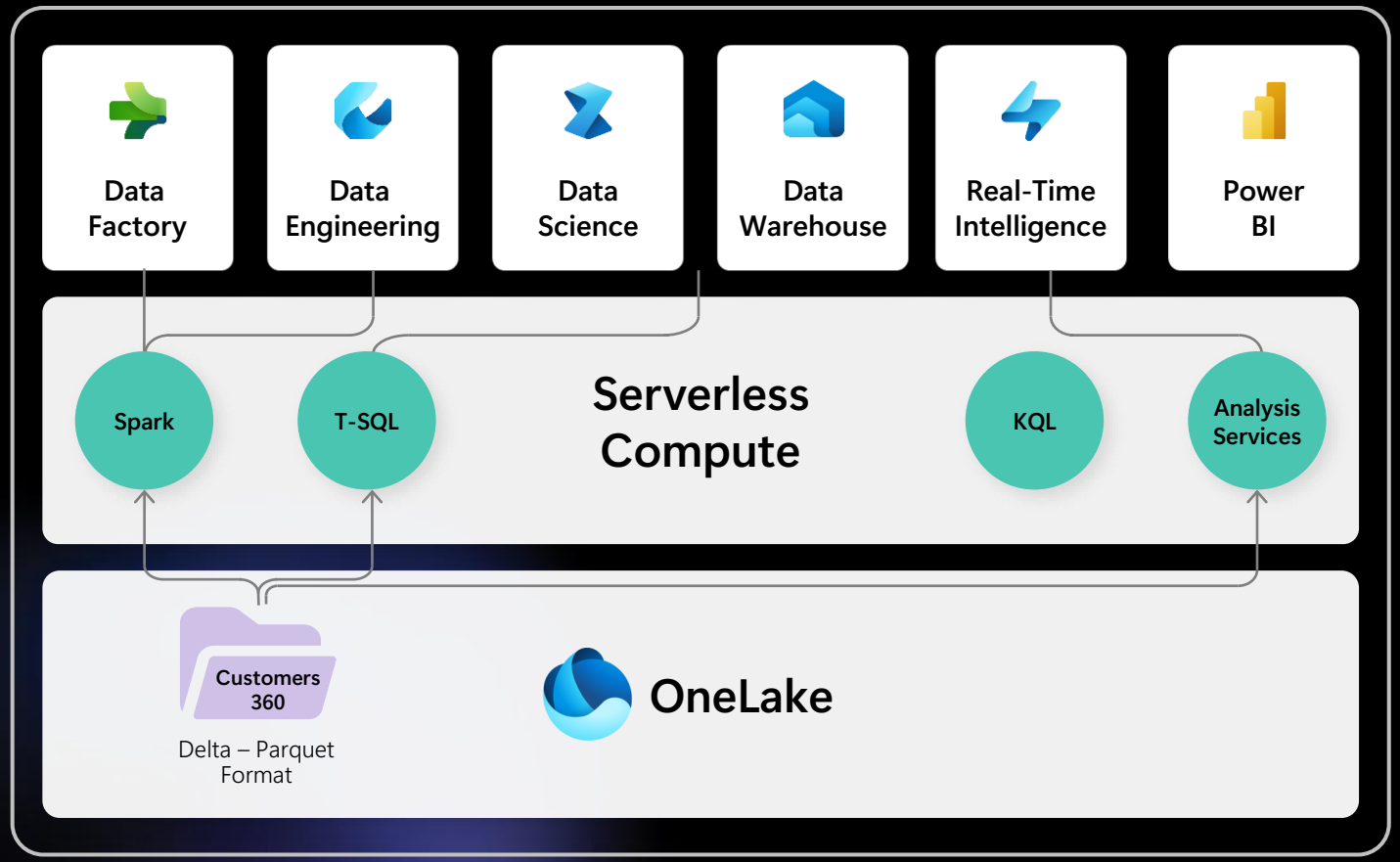
Underlying physical storage is virtualized away.

All storage is zone redundant by default with an option for Geo redundancy.



# One Copy for all computes

**One copy of data can be read by all engines**




Once data is stored in the lake, it is directly accessible by all the engines without needing any import/export.

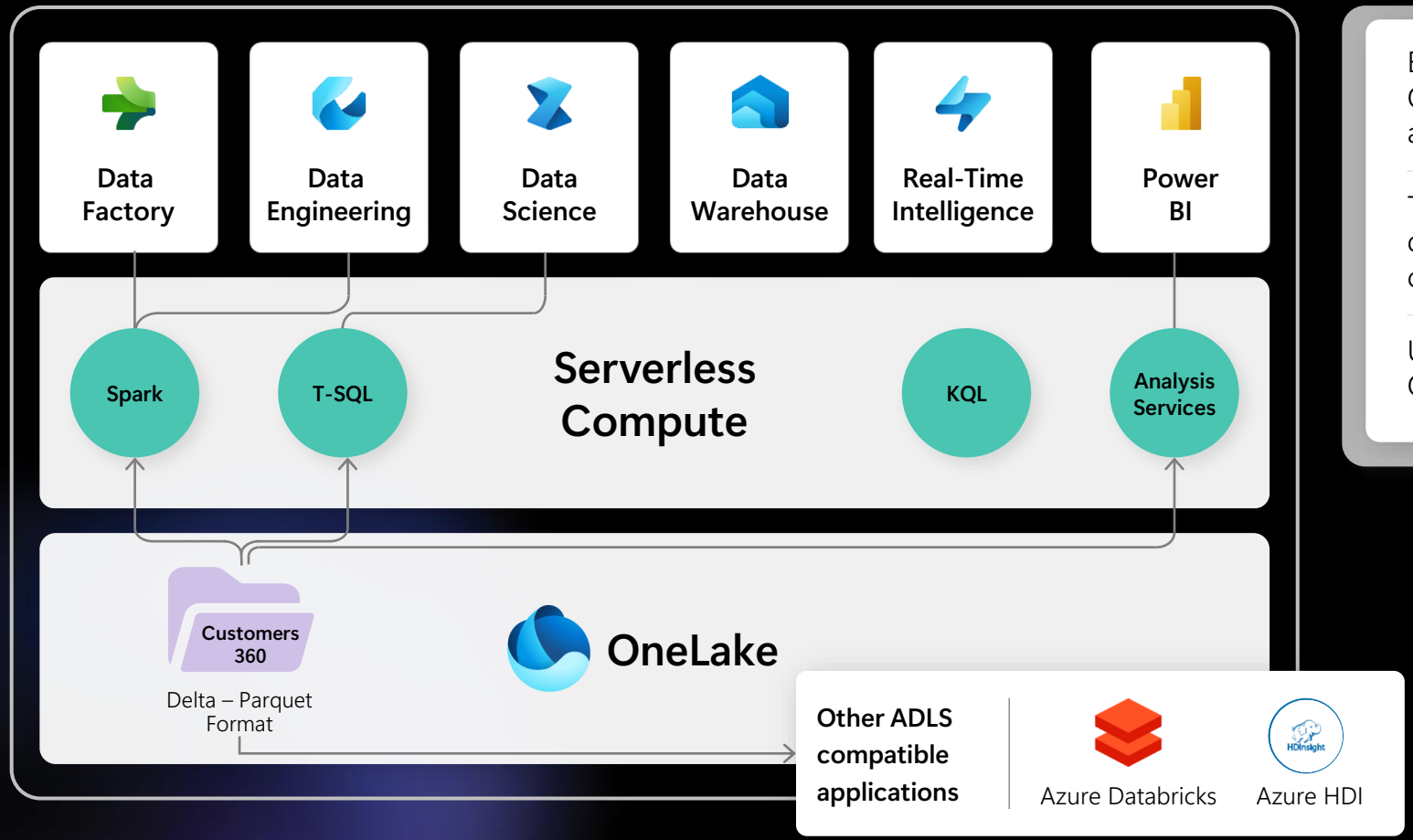
You are able to choose the right engine for the right job.

All the compute engines have been fully optimized to work with Delta Parquet as their native format.

Shared universal security model is enforced across all the engines (coming soon).

# Open Access to data in OneLake

No lock-in with industry standard APIs and open file formats




By supporting the ADLS Gen2 DFS APIs and SDKs, OneLake is compatible with existing ADLS applications.

Tenants will appear as one big storage account with different workspaces appearing as different containers with data organized into folders.

Underlying physical storage is virtualized away. OneLake ensures proper scale and performance.

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# Data Factory



# Data Factory

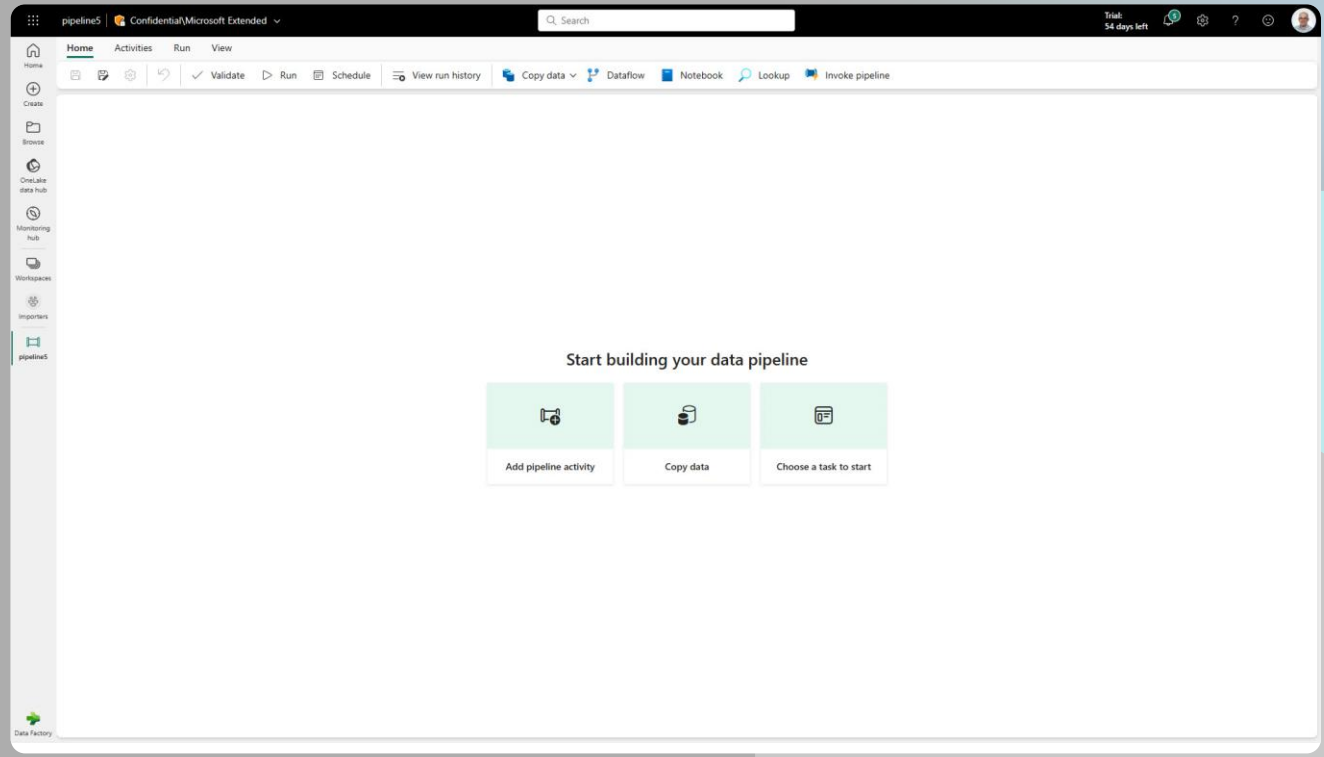
**Data Factory** in Microsoft Fabric provides cloud-scale data movement and data transformation services that allows you to solve the most complex ETL scenarios

Core to Data Factory are Data Pipelines and Dataflows to give users the option to a low-code, collaborative and enterprise scale approach for their ETL process

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***Key Capabilities:***

- Latest capabilities:
- Output destination to Lakehouse
- 170+ connectors available in Data Factory
- Pipeline Lakehouse copy assist
- Create data pipeline in Lakehouse portal
- Pipeline templates
- Pipeline support for Spark notebooks
- Service principle auth support

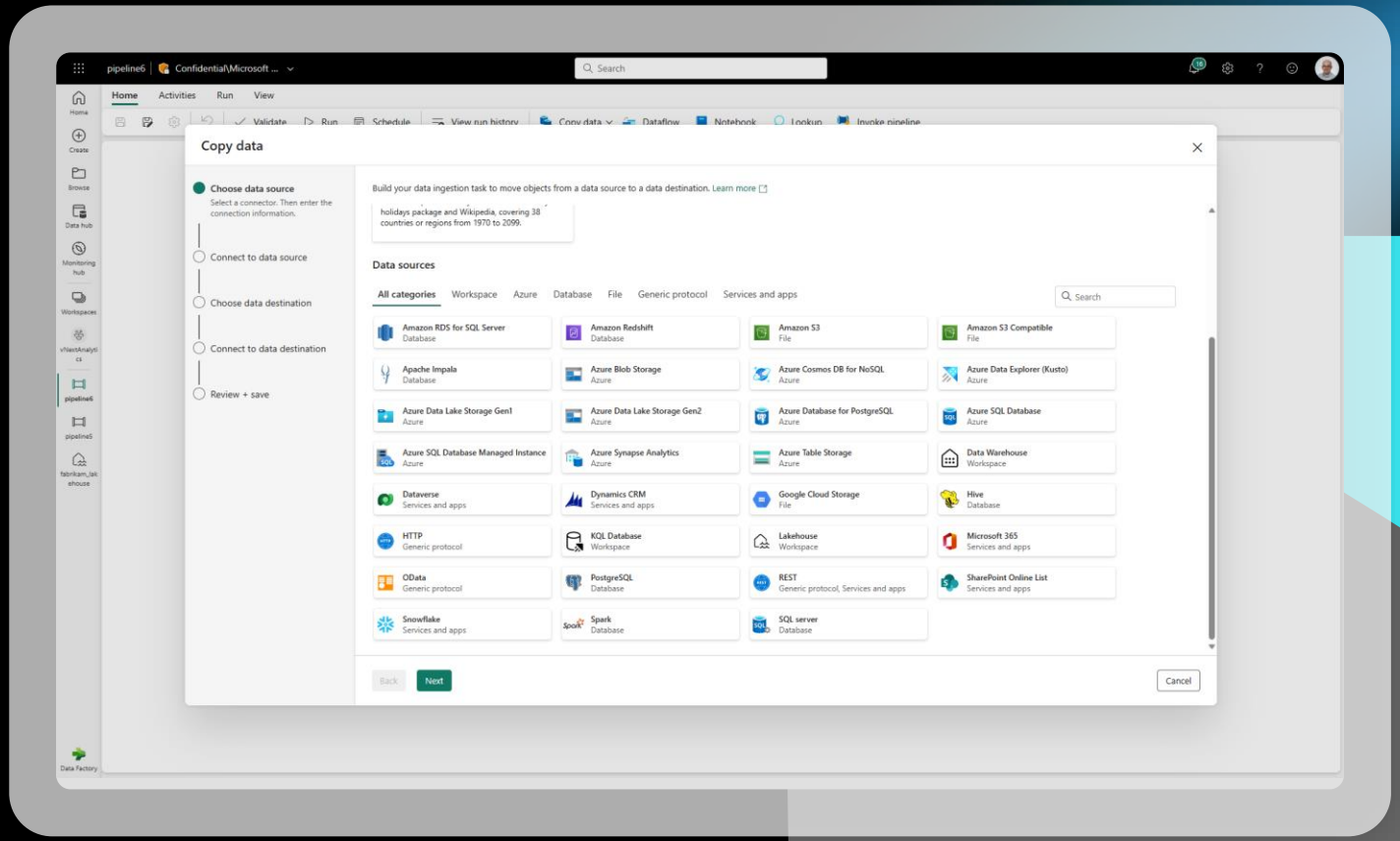



*d help orchestrate, monitor and manage*

# Data Pipelines

**Data Pipelines** enable powerful workflow capabilities at cloud-scale like building complex workflows, moving PB-size data, and defining sophisticated control flow pipelines

Data pipelines can be used to build complex ETL and data factory workflows that can perform a number of different tasks at scale. Additionally, control flow capabilities are built into pipelines so you can build workflow logic which provide loops and conditional



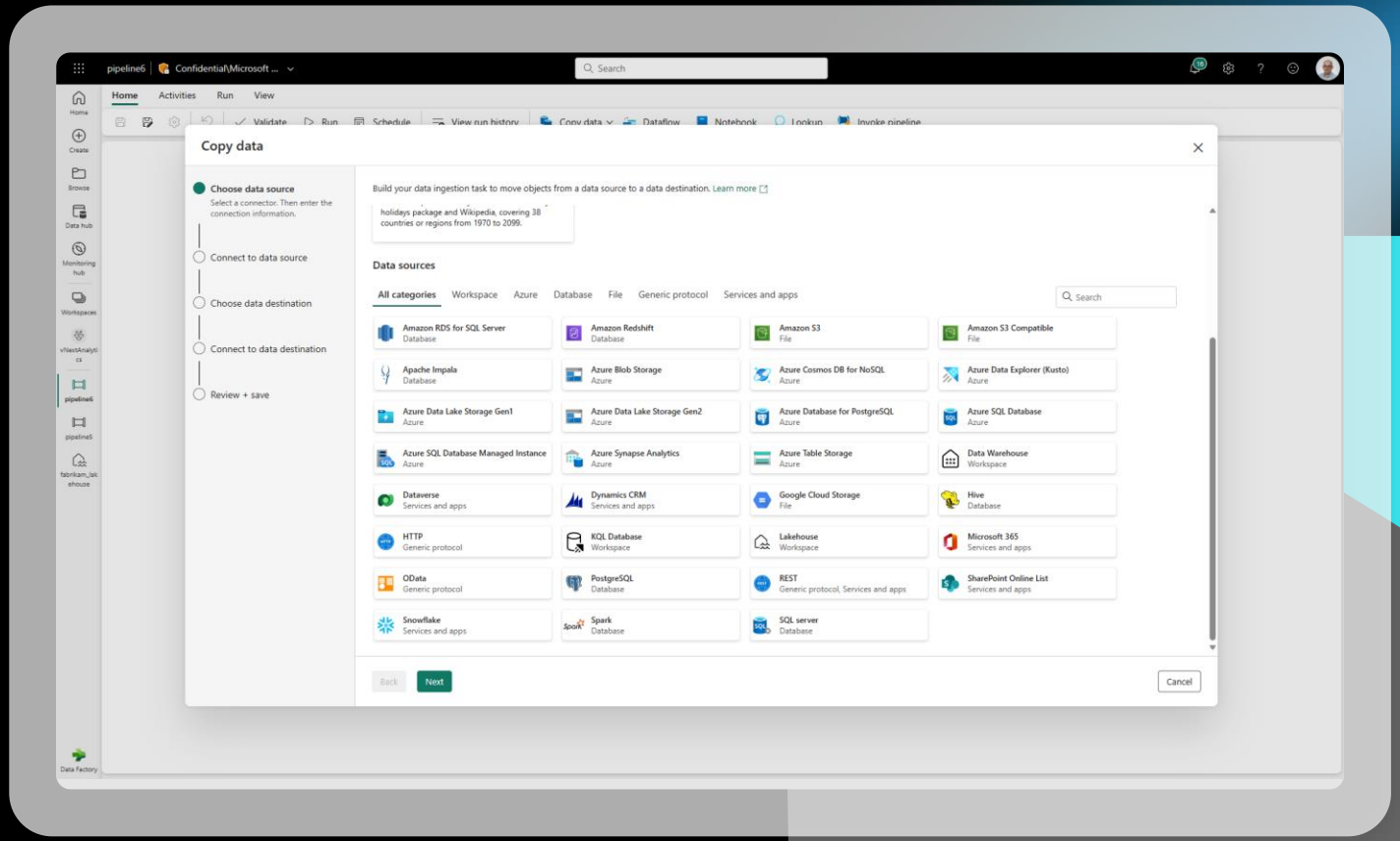
# Data Pipelines | Connectors

**New Connectors** provide a low-code interface for ingesting data from a variety of data sources

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***Connectors:***

- Warehouse Connector; connect to existing Azure
- Lakehouse connector
- 100+ connectors in the copy activity
- Access to on-premises data
- Access protected data inside of a VNET



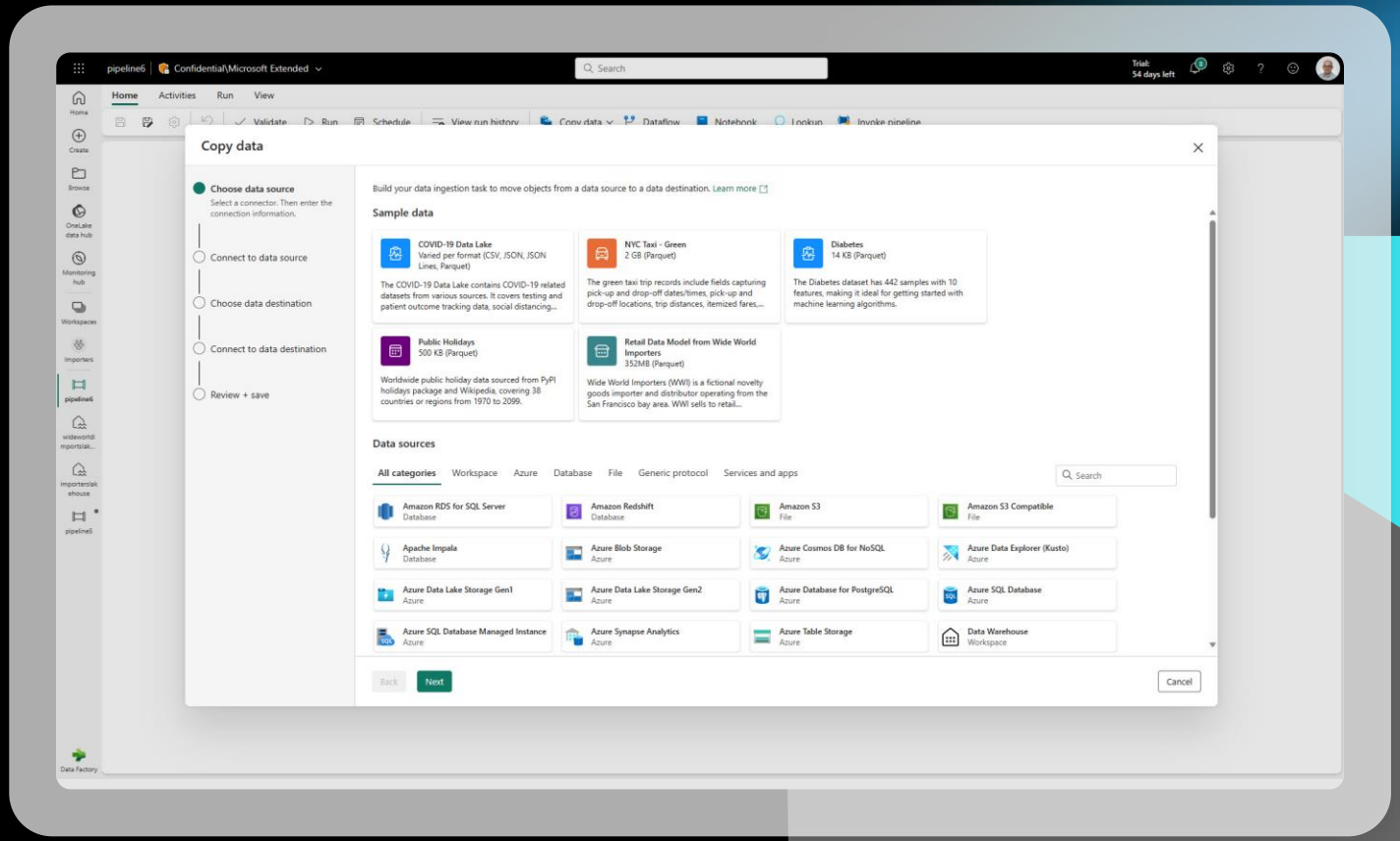
# Data Pipelines | Sample data

**Sample Datasets** helps new users get started quickly, building out their ELT processes using Data Pipelines

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***Sample datasets:***

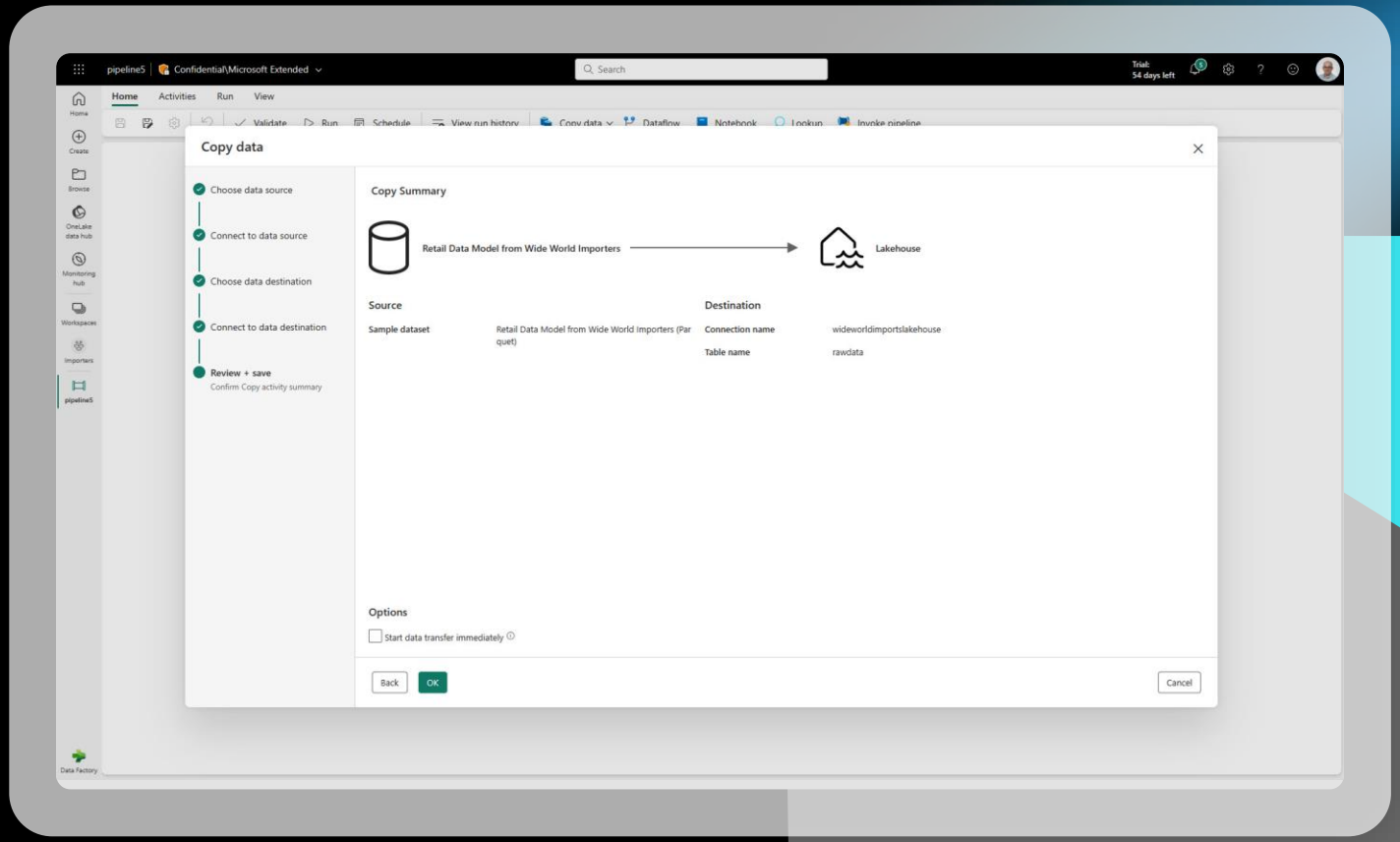
- COVID-19 Data Lake (CSV, JSON, JSON Lines, Parquet)
- NYC Tax – Green (2GB Parquet)
- Diabetes (14K Parquet)
- Public Holidays (500KB Parquet)
- Retail Data Model from Wide World Importers (352MB Parquet)



# Data Pipelines | Lakehouse copy assist

**Simply** copying data to a Lakehouse with copy assist capabilities within the Data Pipeline

Additionally, users can create a Data Pipeline without having to leave the Lakehouse portal



# Data Pipelines | Templates

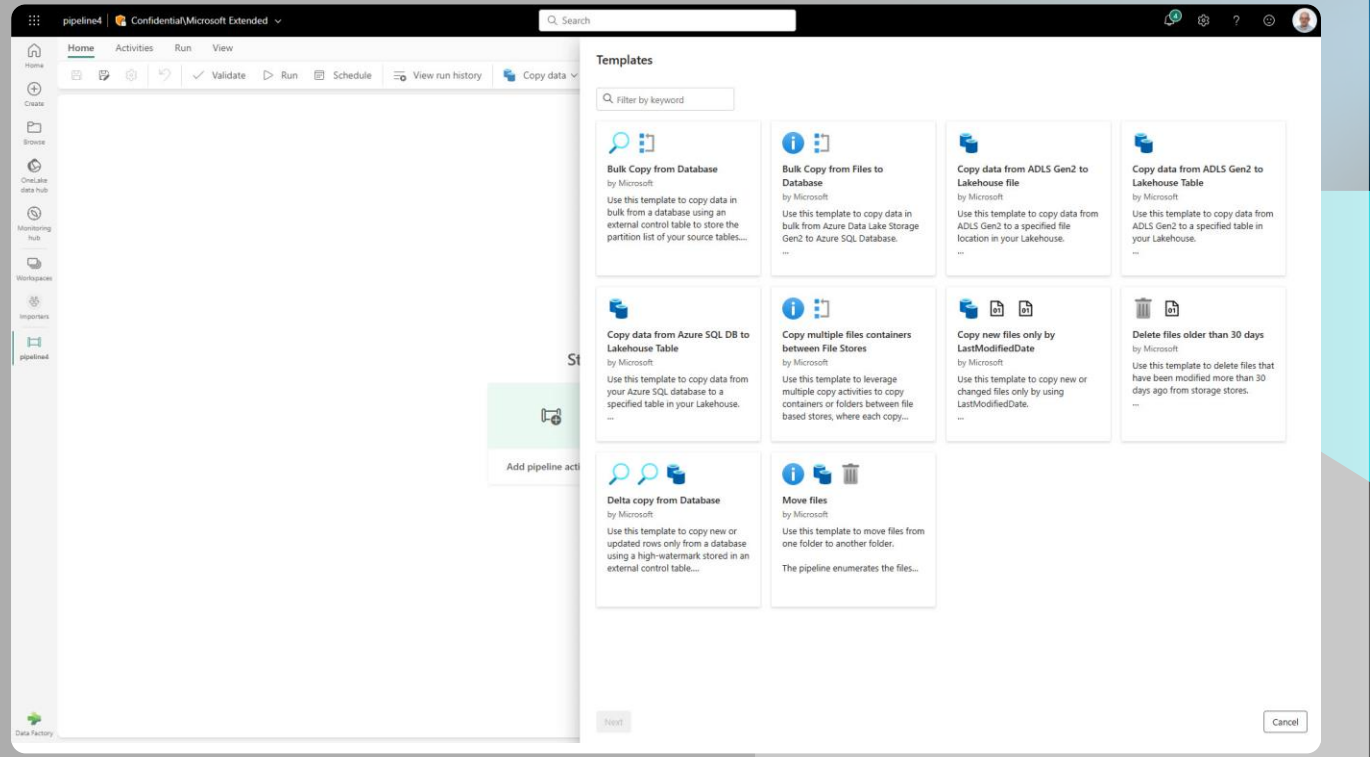
**Quickly** get started with data integration

Template help reduce development time by providing an easy way to create pipeline for common data integration scenarios

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***Available Data Pipeline Templates:***

- Bulk copy from Database
- Bulk copy from File to Database
- Copy data from ADLS Gen2 to Lakehouse file
- Copy from ADLS Gen2 to Lakehouse Table
- Copy data from Azure AQL DB to Lakehouse Table
- Copy multiple files containers between File Stores
- Copy new files only by Last Modified Date
- Delete files older than 30 days
- Delta copy from Database
- Move files



# Dataflow

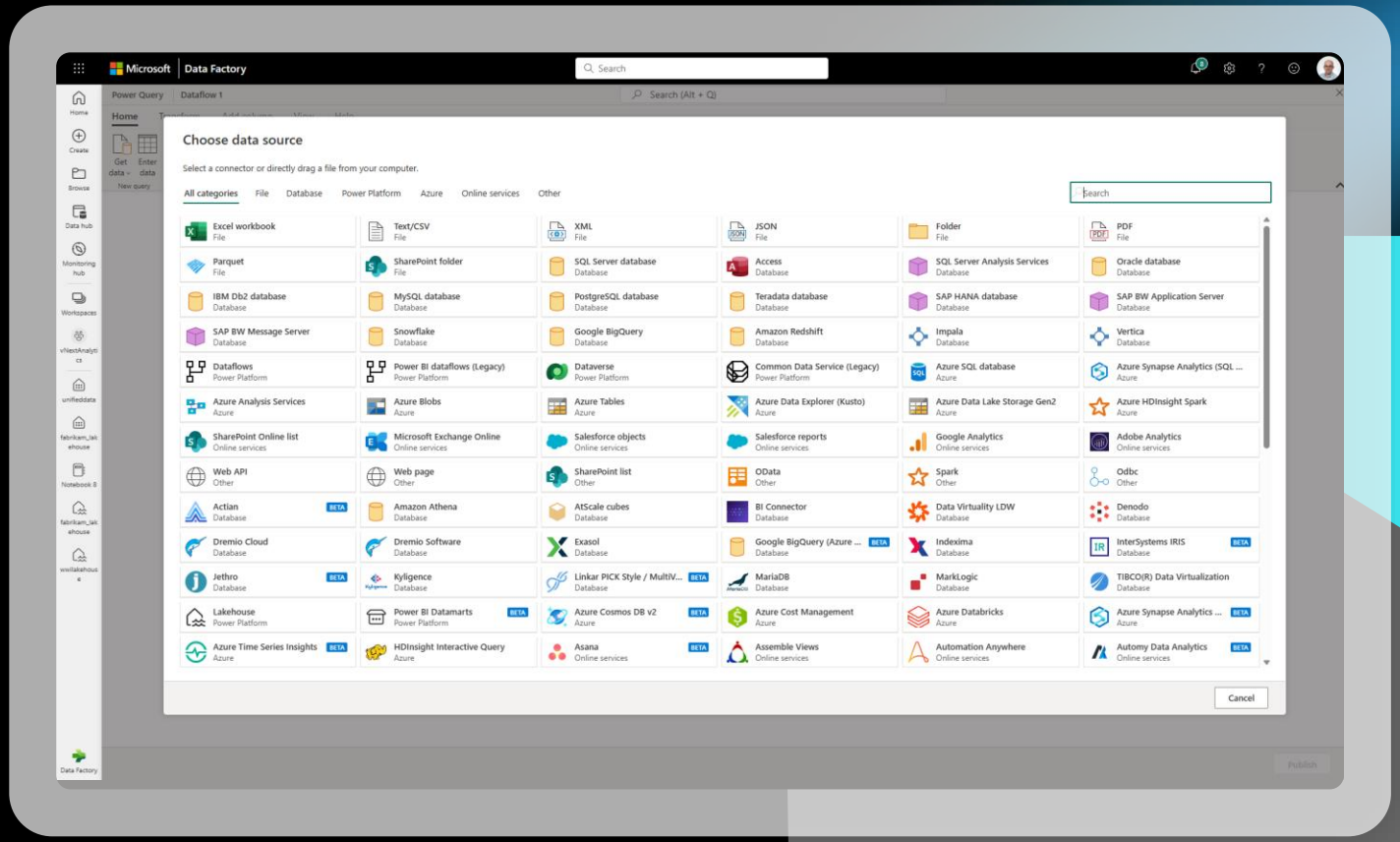
**Dataflow** provides a low-code interface for ingesting data from hundreds of data sources

Dataflow quickly and easily unify disparate data sources, establish a more collaborative analytics approach, and promote more informed, agile decision making.

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***Key Capabilities:***

- Accelerate data transformation with code-free data flows
- Scale out using Fabric compute and Data Factory fast copy
- Load results of data transformations into multiple destinations (Azure SQL Databases, Lakehouse, etc.)

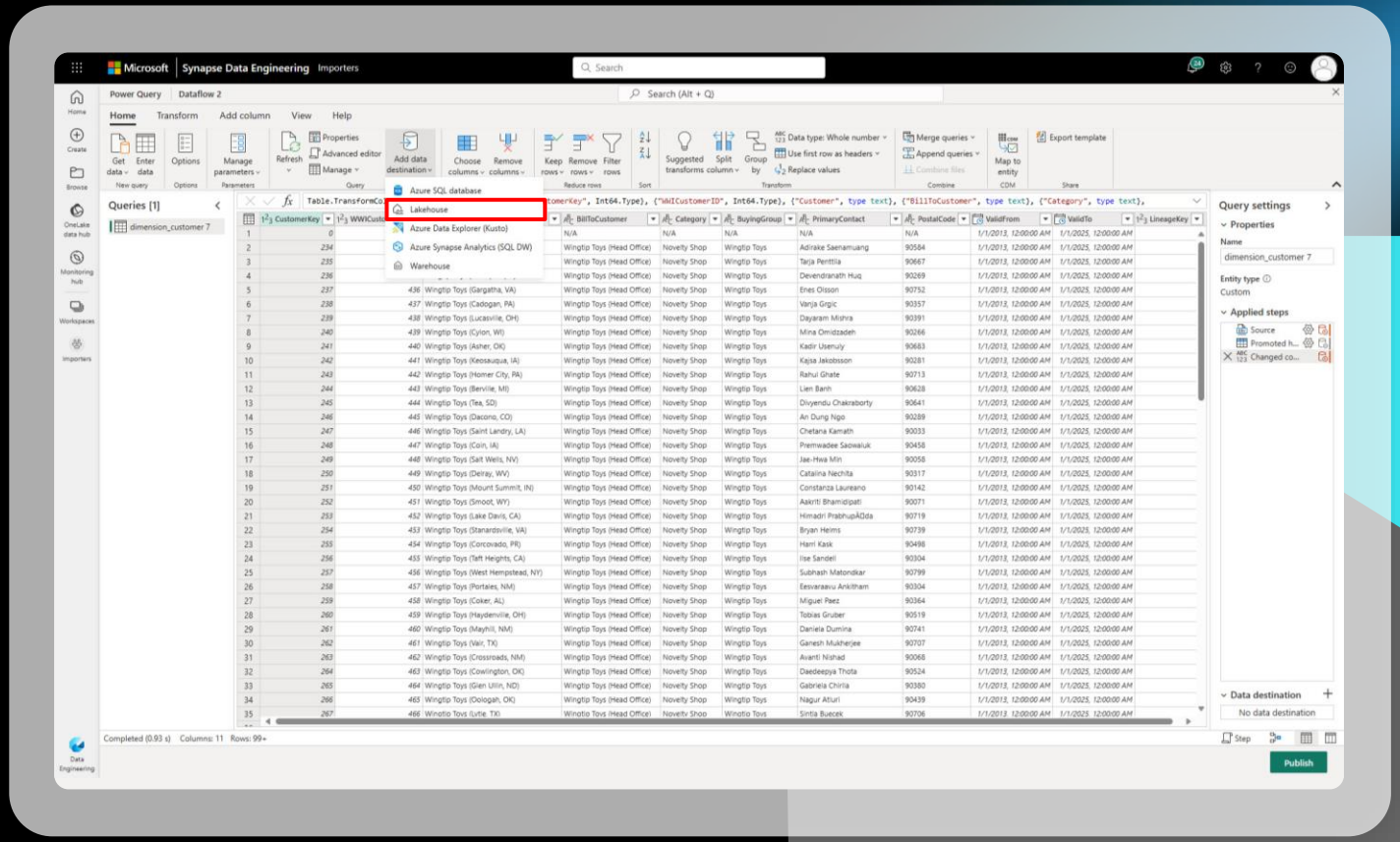


# Dataflow | Output to Lakehouse

**Simply** write into a Lakehouse from a Dataflow

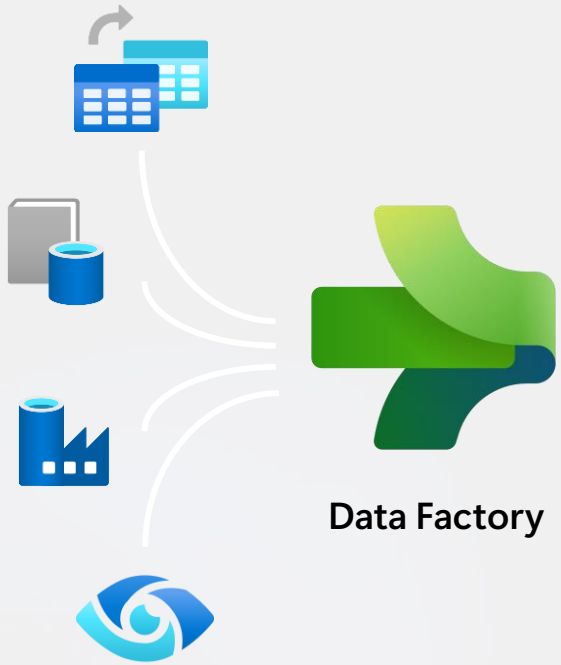
Users select the Lakehouse output destination from the list and configure the connection.

This requires the Lakehouse connector to be installed as a custom connector into your data gateway when loading data from on-premise



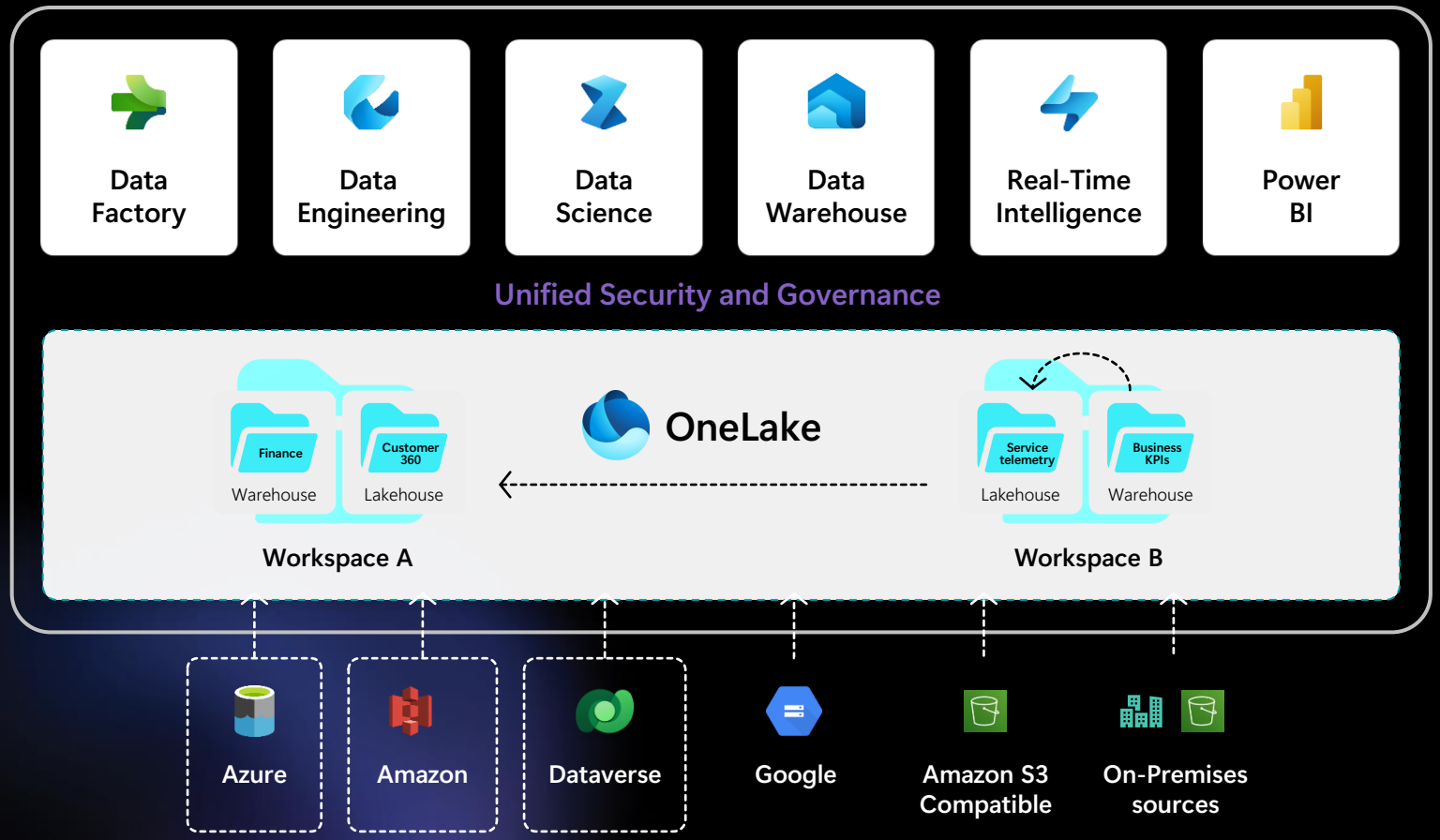
# Unifying data in OneLake

## Seamlessly connect to more than 170+ data stores




| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Azure Database for PostgreSQL | Azure Databricks Delta Lake | Amazon RDS for Oracle | Amazon RDS for SQL Server | Amazon Redshift | Phoenix | PostgreSQL | Presto | Magento (Preview) |
| Azure SQL Database | Azure SQL Database Managed Instance | Apache Impala | Azure SQL Database Managed Instance | DB2 | SAP BW Open Hub | SAP BW via MDX | SAP HANA | Oracle Eloqua (Preview) |
| Azure Table Storage | MongoDB Atlas | Drill | Google AdWords | Google BigQuery | SAP Table | SQL server | Spark | PayPal (Preview) |
| Azure Cosmos DB (MongoDB API) | Azure Cosmos DB (SQLAPI) | Greenplum | HBase | Hive | Amazon S3 | Amazon S3 Compatible | FTP | SAP Cloud For Customer |
| Azure Data Lake Storage Gen1 | Azure Data Lake Storage Gen1 for Cosmos Structured Stream | Informix | MariaDB | Microsoft Access | File system | Google Cloud Storage (S3APD) | HDFS | Salesforce Marketing Cloud |
| Azure Data Lake Storage Gen1 for Cosmos Structured Stream | Azure Database for MariaDB | MySQL | Netezza | Oracle | HTTP | Oracle Cloud Storage (S3AP) | SFTP | Shopify (Preview) |
| Teradata | Vertica | ODBC | OData | REST | Amazon Marketplace Web Service | Concur (Preview) | Dataverse (Common Data Service for App) | Web Table |
| Jira | Kusto | SharePoint Online List | Dynamics 365 | Dynamics AX | Dynamics CRM | Cassandra | Couchbase (Preview) | MongoDB |

# Shortcuts virtualize data across domains and clouds




A shortcut is a symbolic link which points from one data location to another.

Create a shortcut to make data from a warehouse part of your lakehouse.

Create a shortcut within Fabric to consolidate data across items or workspaces without changing the ownership of the data. Data can be reused multiple times without data duplication.

Existing ADLS Gen2 storage accounts and Amazon S3 buckets can be managed externally to Fabric and Microsoft while still being virtualized into OneLake with shortcuts.

All data is mapped to a unified namespace and can be accessed using the same APIs including the ADLS Gen2 DFS APIs.

# Dataverse makes it easy to connect Power Platform and Dynamics 365 to Fabric

- No Copy. No ETL.
- Direct Connection via Dataverse
- Insights democratized to all low code apps and business using Fabrics 7 core workloads
- Makers informed by insights improves quality of applications
- Data is governed



Microsoft Power Platform



Microsoft Dataverse



Microsoft Dynamics 365



Microsoft Fabric

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# Data Engineering

# Lakehouse | Overview

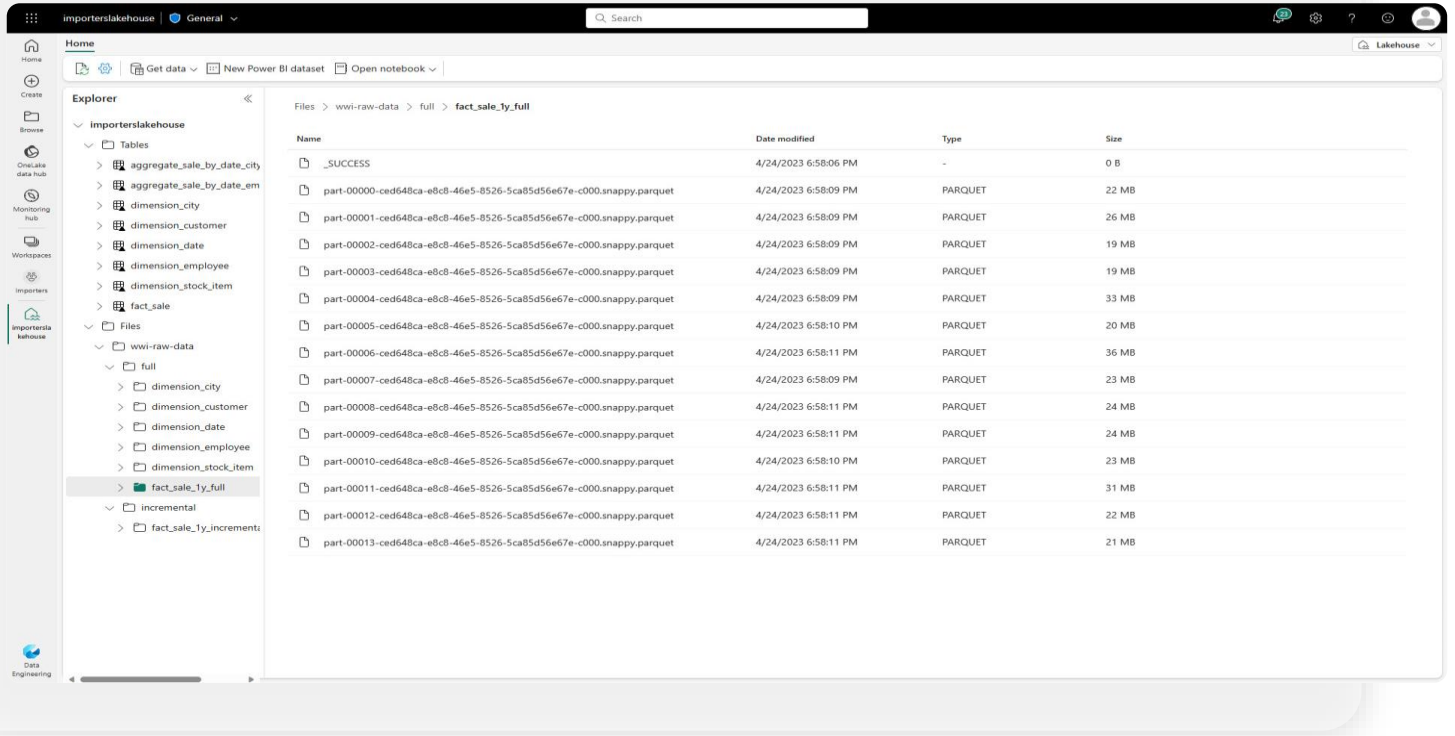
Store, manage and analyze all your data in a single location & easily share across the entire enterprise

Quickly and easily create a Lakehouse without having to provision and configure compute, storage and networking

---

***Key Capabilities:***

- Flexible and scalable solution that enables organizations to handle large data volumes of all types and sizes
- Built-in SQL endpoint unlocks data warehouse capabilities on top of your Lakehouse with no data movement
- Use ‘direct lake’ mode to build reports in seconds directly on top of the data lake with blazing fast performance
- Easily ingest data into the Lakehouse through a variety of methods
- Share your Lakehouse as a data product with consumers



# Lakehouse | Overview

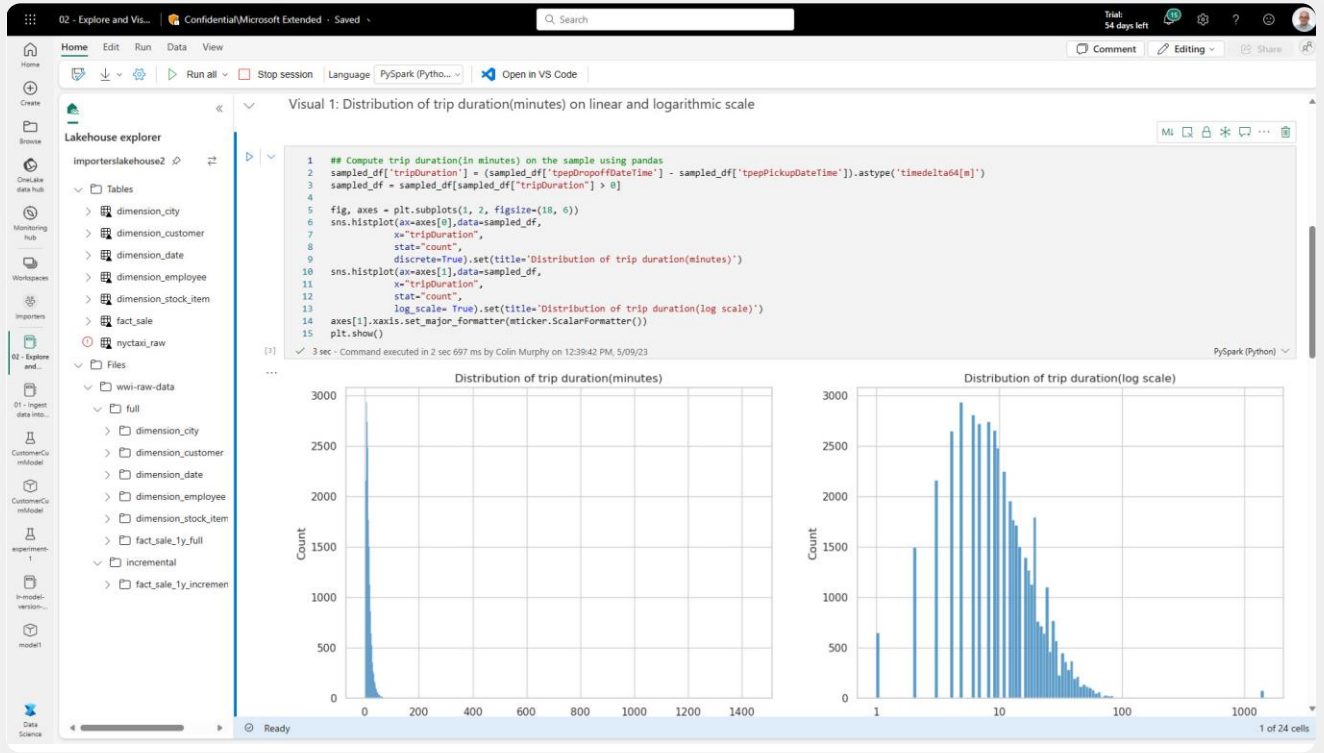
Immersive authoring experience for data developers

Rich notebook capabilities including native Lakehouse integration, real-time collaboration with commenting support, auto-save support, lightweight scheduling and pipeline integration

---

***Key Capabilities:***

- Manage your Python and R libraries through in-line installs using commands like %pip install
- Advanced notebook development support with ability to reference notebooks in notebooks, and snapshots for easy troubleshooting
- In context monitoring complete with real time advice and error analysis
- Streamline data prep without giving up the power of reproducibility of Python



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# Data Warehouse



# Data Warehouse | Overview

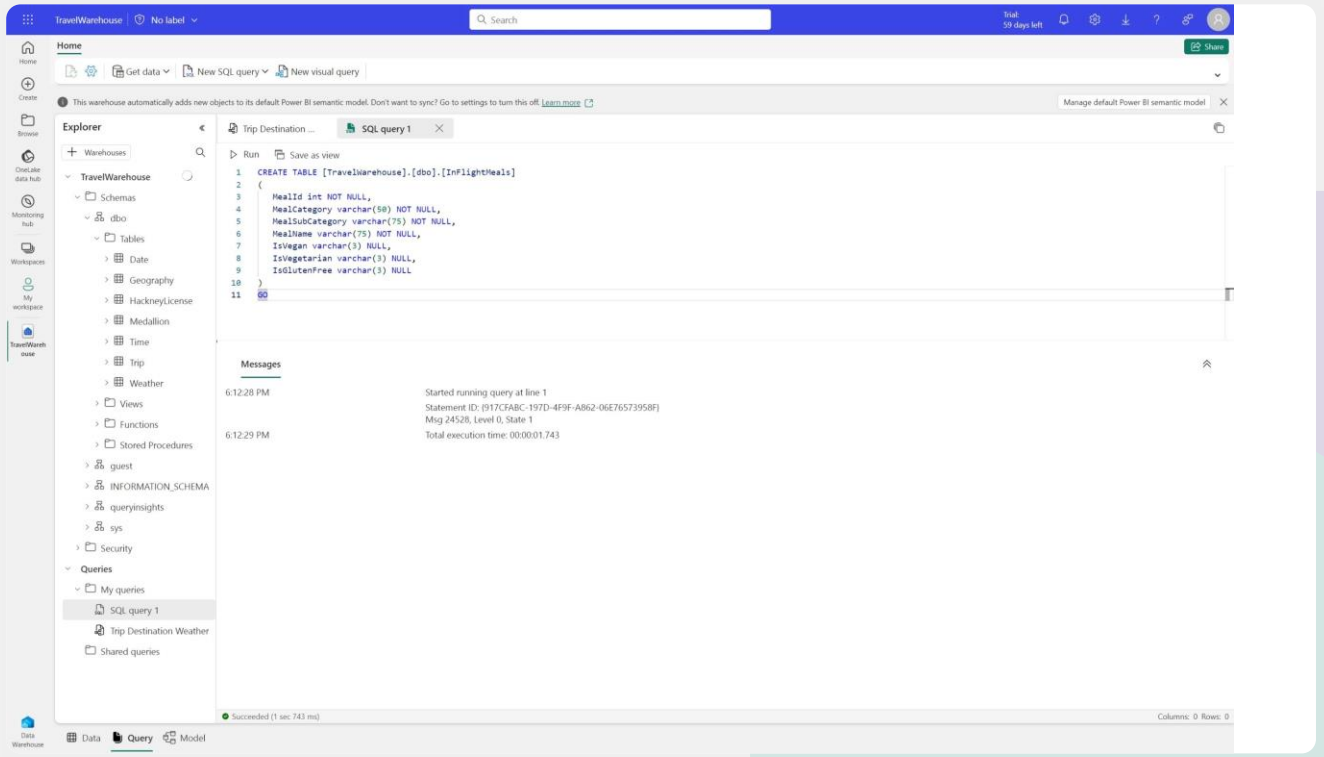
Enterprise scale data warehouse with open standard format

No knobs performance with minimal set-up and deployment, no configuration of compute or storage needed

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***Key Capabilities:***

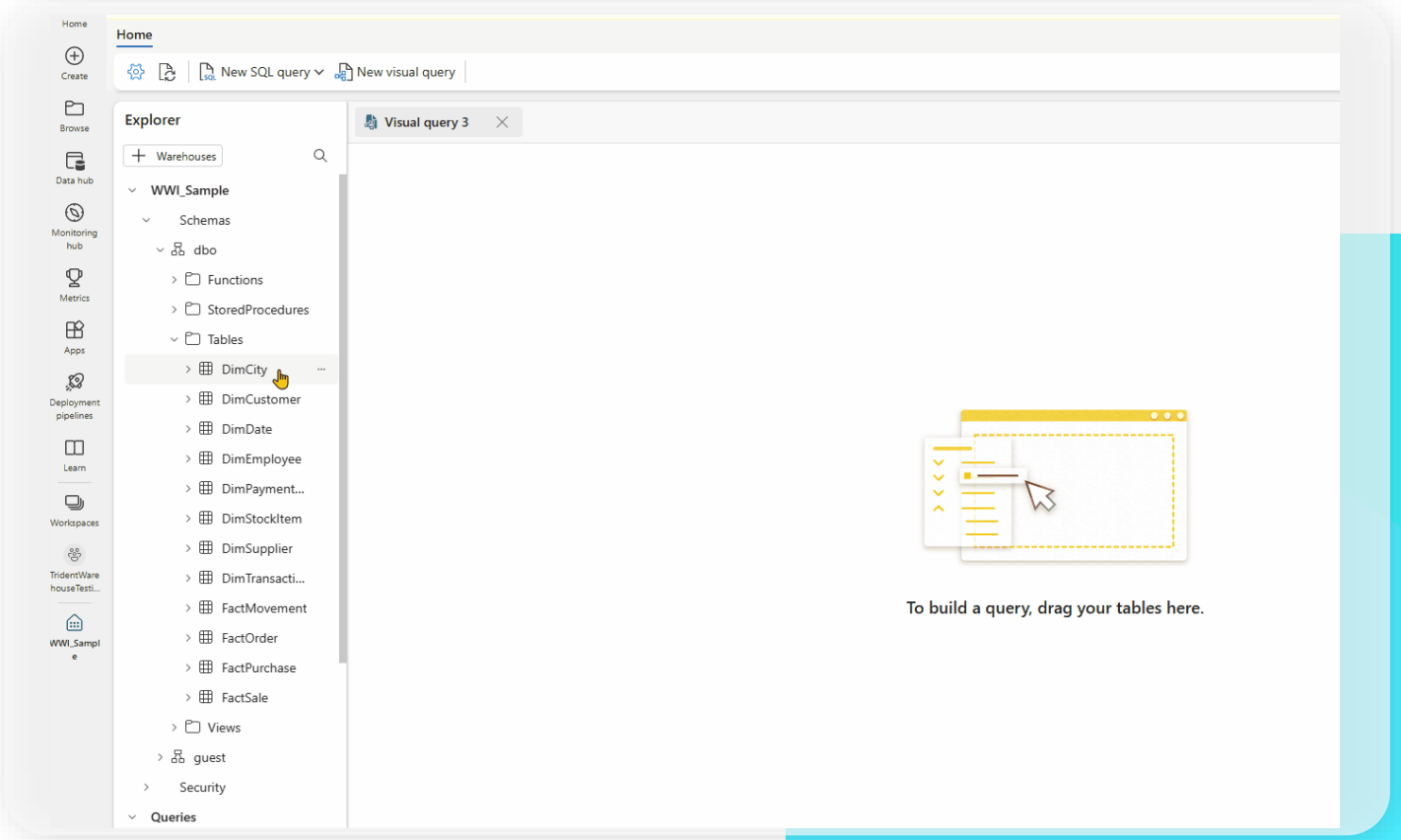
- Lake-centric warehouse stores data in OneLake in open Delta format with easy data recovery and management
- Use Fabric Mirroring for Zero-ETL integration of data from Azure SQL, Snowflake, or Azure Cosmos DB
- Data loading and transforms at scale, with full multi-table transactional guarantees provided by the SQL engine
- Virtual warehouses with cross-database querying and a fully integrated semantic layer
- Flexibility to build data warehouse or data mesh based on organizational needs and choice of no-code, low-code, or T-SQL for transformations



# Visual Query Editor

***Key Capabilities:***

- Easy and efficient method to transform data
- No-code experience
- Interface like Power Query
- No data duplication
- Query executed on every call
- Direct Lake not available



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# Fully integrated Power BI semantic layer

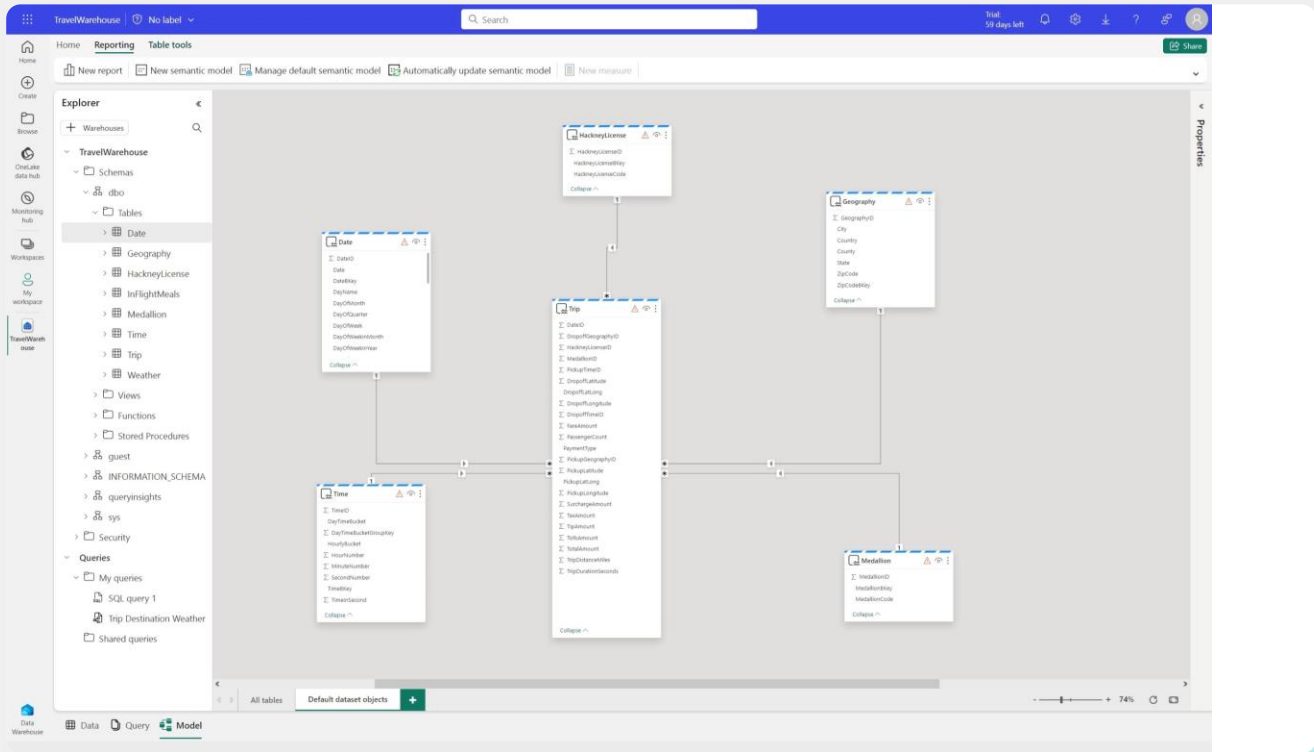
Reduce integration and gain insight from your data in seconds

Built-in Power BI enables everyone to visualize their data in seconds.

---

***Key Capabilities:***

- Auto-generated semantic models always in sync
- Default dataset in Direct Lake mode but automatically switches to Direct Query or Import mode as security or performance needs change
- Flexibility to add/remove tables to dataset
- Full web authoring experience for creating measures



# Secure by default

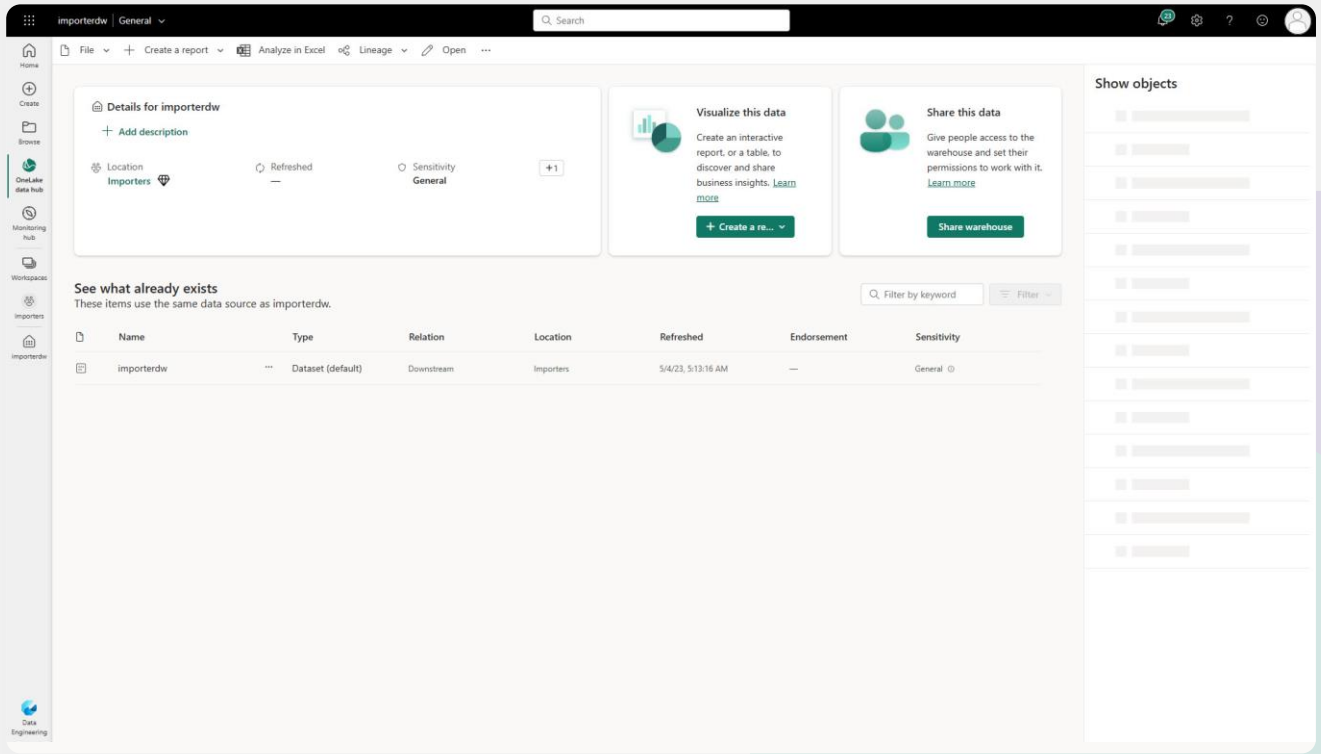
Keep data secure for any role accessing it and ensure peace of mind

Customers can secure their data using familiar constructs and ensure data is only visible to those authorized to do so

---

***Key Capabilities:***

- **Workspace roles:** Workspace roles are used for collaboration with team. Add users to workspace with role assignment of Admin, Member, Contributor, Viewer
- **Artifact permissions:** Artifact permissions are used for sharing for consumption of Warehouses. Provide access and share individual Warehouses with specific permissions
- **Data security:** Use T-SQL, GRANT, REVOKE or DENY to secure any object within Warehouse. Users can be assigned to built-in custom roles.
- **Sensitivity labels:** Apply sensitivity labels on your Warehouse to classify sensitive data.
- **Granular security:** Implement row or column security or Dynamic Data Masking for granular data access control



# Mirroring in Microsoft Fabric

## Simplify near real-time intelligence

Fabric Mirroring enables adding existing databases and data warehouses to Fabric without any ETL.

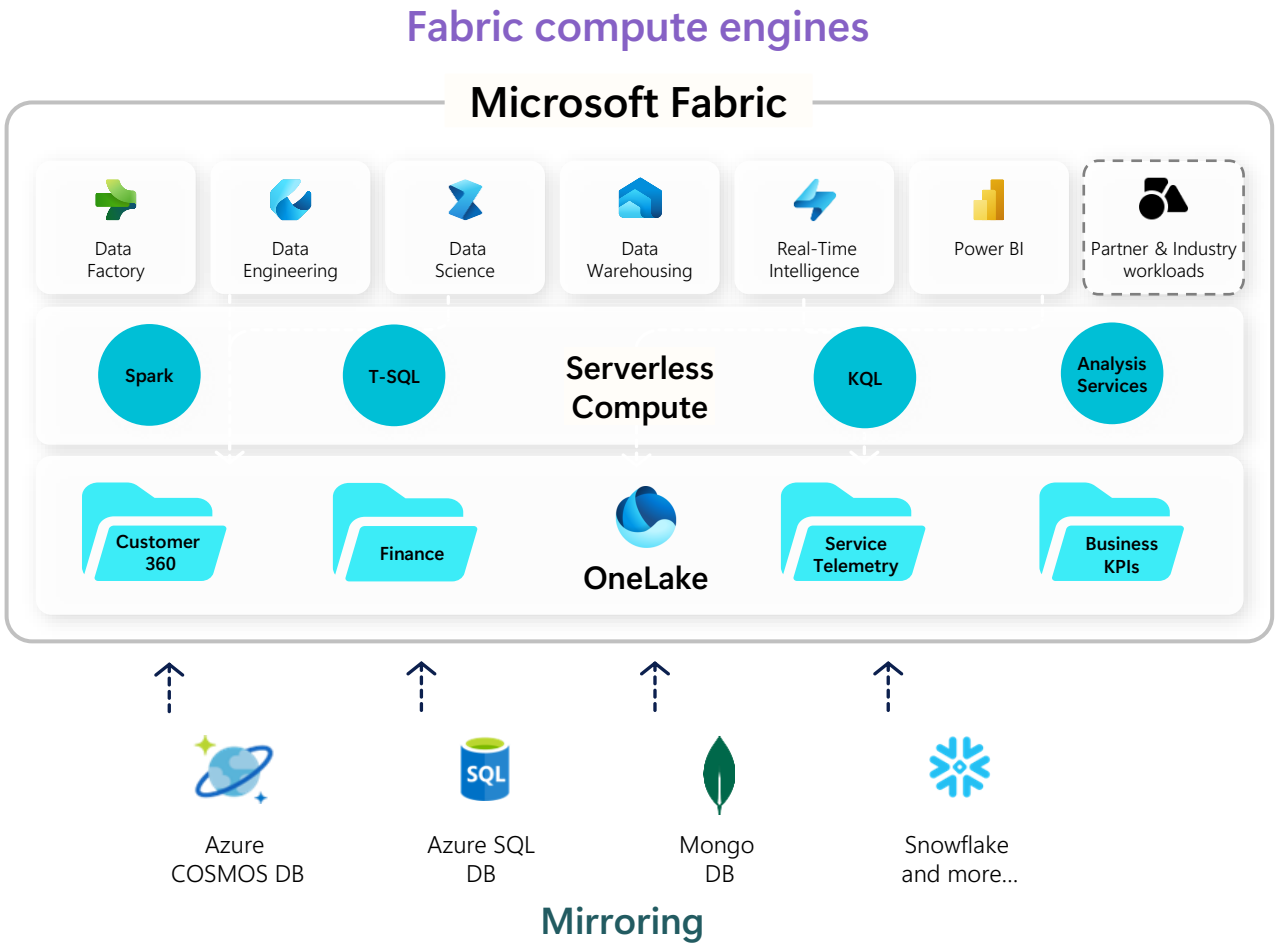
A full editing experience of the source database is available for the Mirrored database.

Data is replicated into OneLake in Delta format and kept up-to-date in near-real-time.

All the Fabric experiences instantly work with the OneLake replica.

Analysts and Data Scientists can work with real-time data.

The replica protects operational databases from analytical queries.



# Thank You!!

- Contact:
  - Email: russ.seaman@tdsynnex.com
  - LinkedIn: www.linkedin.com/in/russseaman

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# Q&A

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